MPCA Project

TITLE OF THE PROJECT:

Automatic hand sanitizer dispenser with counter

Team Members: AIML SECTION H

Sl. No	Name	SRN
1	AMULYA P	PES1UG24AM458
2	ANAGHA KAUSHIK	PES1UG24AM459
3	THRUPTHI G D	PES1UG24AM476

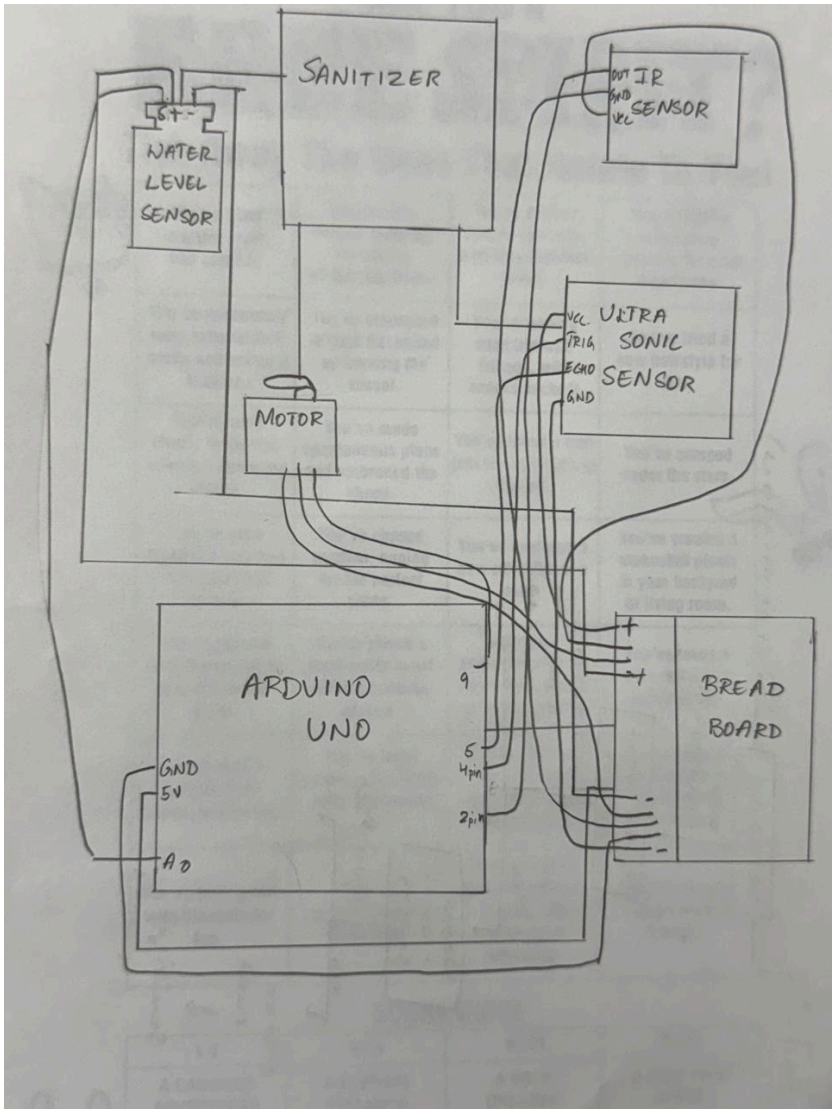
Components used:

Arduino IDE, Arduino Uno, IR Sensor, Ultrasonic And Level Sensor, Mini Water Pump, Motor Driver, Jumper Wires, Breadboard,Battery.

Explanation about the project :

Our project detects a user's hand using an IR sensor and dispenses sanitizer while maintaining a count of the number of uses. Once the IR sensor is triggered it in turn triggers the Ultrasonic sensor which will sense if the user's hand is within an optimal range of distance. The Ultra sonic sensor then triggers the water level sensor which will check if the level of sanitizer is above a certain level. Once all these criterias pass, the servo motor runs and pulls the pump down which dispenses sanitizer

Circuit Diagram:



Pseudocode:

```
START

INITIALIZE:
   Set IR_SENSOR as INPUT
   Set TRIG as OUTPUT
   Set ECHO as INPUT
   Attach servo to SERVO_PIN
   Set servo position to 150 (initial position)
   Initialize count = 0
   Start serial communication

LOOP FOREVER:

   READ ir value from IR_SENSOR

   // Ultrasonic distance measurement
   Set TRIG LOW
   Wait 2 microseconds
   Set TRIG HIGH
   Wait 10 microseconds
   Set TRIG LOW

   Measure duration from ECHO
   Calculate distance = (duration * 0.034) / 2

   // Water level reading
   Read level from WATER sensor

   // Display values
   Print IR value
   Print distance
   Print water level
   Print count

   // Main condition check
   IF (ir == LOW AND distance > 2 AND distance < 12) THEN
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        IF (level < 300) THEN
            Print "Low Liquid Warning"
        ENDIF

        Move servo to 60 (press action)
        Wait 1200 milliseconds

        Move servo to 150 (release action)
        Wait 500 milliseconds

        Increment count by 1

    ENDIF

    Wait 500 milliseconds

END LOOP

END
```
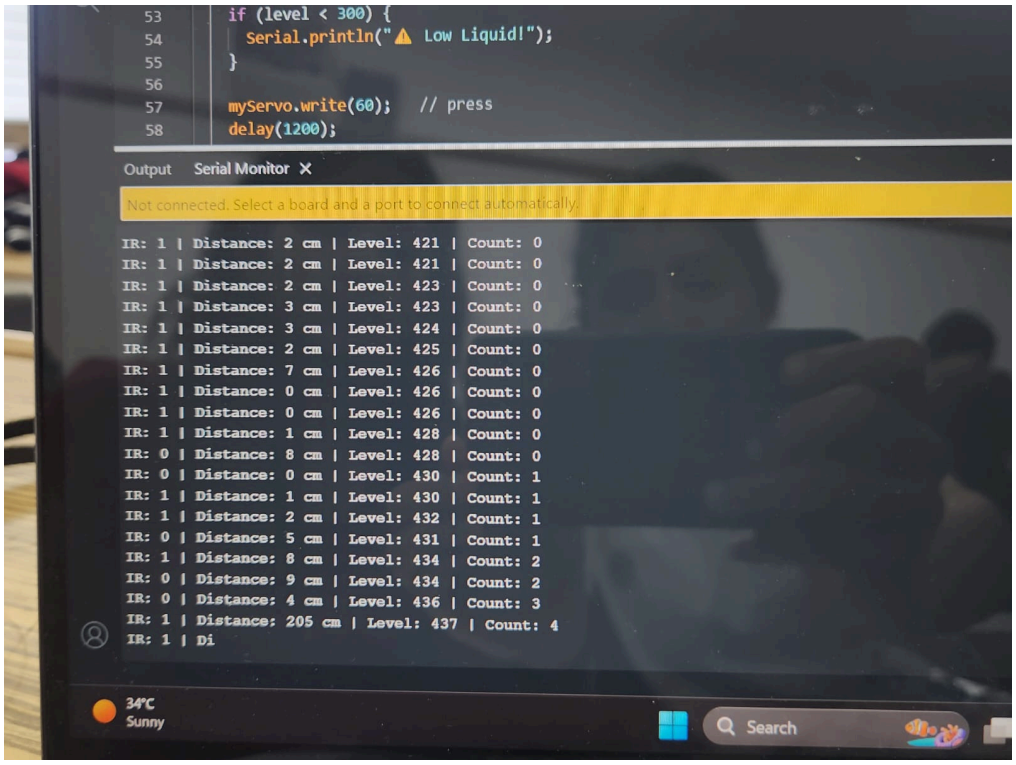

Output Pictures with explanations:



- The IR value is either 0 or 1 where 1 means no hand detected and 0 means a hand detected.
- The valid range for ultra sonic sensor is 2-12 cm.
- The water level sensor checks if the quantity is above 300.
- Count tells the number of times the servo pressed i.e, number of times the dispenser was activated.
- As we can see in line 15, all the criterias were met so the count increased to 1 indicating the dispenser was pumped once.
- Similarly, each time the criterias are met the count increases.